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ZYNEX CertVerify

Cryptographic - Blockchain Certificate Verification System

- React
- Node.js
- Python Flask
- MongoDB

"Eliminating certificate fraud with steganography, blockchain, and Gemini AI"

TEAM- MNM's

THE PROBLEM

Certificate fraud is a global epidemic



41%

of employers have caught candidates submitting fake degrees **(HireRight, 2023)**



Billions in Costs

Fake certificates cost companies billions in bad hires and legal liability



Days to Weeks

Manual verification takes days to weeks — no instant employer tool exists

Why Current Solutions Fail

Email Verification

Slow, error-prone process

Watermarks

Easily removed or replicated

No AI Forensics

No pixel-level edit detection

No Cryptographic Binding

No link between image and data

PROBLEM SOLVING EFFECTIVENESS

OUR SOLUTION

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6 independent security layers — any tampering causes 3+ to fail

CERTIFICATE ISSUED →

01 SHA-256 Hash →

02 AES-256 Encrypt →

03 LSB Steganography →

04 RSA-2048 Signature →

05 QR Code (JWT Token) →

06 Blockchain Ledger →

07 Gemini AI Vision

🔐 Cryptographic Fingerprint

SHA-256 hash of student data creates unique, irreversible identifier

🛡️ Military-Grade Encryption

AES-256 encrypts payload before hiding — medical/legal standard

👁️ Invisible Payload

LSB steganography hides data in image pixels — image looks identical

✍️ Institutional Authority

RSA-2048 digital signature proves certificate issued by valid institution

📱 One-Click Verification

QR code contains JWT token for instant employer scan and validation

⛓️ Immutable Record

Blockchain ledger creates tamper-proof proof-of-issuance with timestamp

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FUNCTIONAL MVP

13 FEATURES

Feature	Status	Technology
Institution login + role-based access	✓ LIVE	JWT + bcrypt (12 rounds)
Multi-step certificate system	✓ LIVE	React + Form validation
LSB steganography embed	✓ LIVE	Python Flask + NumPy
SHA-256 + Double SHA-256 hashing	✓ LIVE	node-forge crypto
RSA-2048 digital signature	✓ LIVE	node-forge RSA
AES-256 payload encryption	✓ LIVE	node-forge AES
QR code generation (JWT-signed)	✓ LIVE	qrcode library
Proof-of-work blockchain ledger	✓ LIVE	Custom Node.js PoW
6-layer employer verification	✓ LIVE	React + Express API
Gemini AI Vision forensic analysis	✓ LIVE	Google Gemini 2.0 Flash
LSB audit log + student/user refs	✓ LIVE	MongoDB + Mongoose
Python server real-time health badge	✓ LIVE	Health check endpoint
Admin dashboard with stats + charts	✓ LIVE	Recharts + React

USER EXPERIENCE & DEMONSTRATION

THREE VERIFICATION FLOWS

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Cyberpunk glassmorphism — dark, premium, technical aesthetic

🏛️ Institution Flow

1. Login with credentials
↓
2. Issue Certificate (fill form)
↓
3. Auto: Hash + Sign + Embed
↓
4. Blockchain + QR generated
↓
5. Download secured PNG

🔍 Employer Flow

1. Enter Certificate ID
↓
2. Upload stego image
↓
3. 6-layer security check
↓
4. AI forensic analysis
↓
5. Verdict: GENUINE / FAKE

📱 QR Code Flow



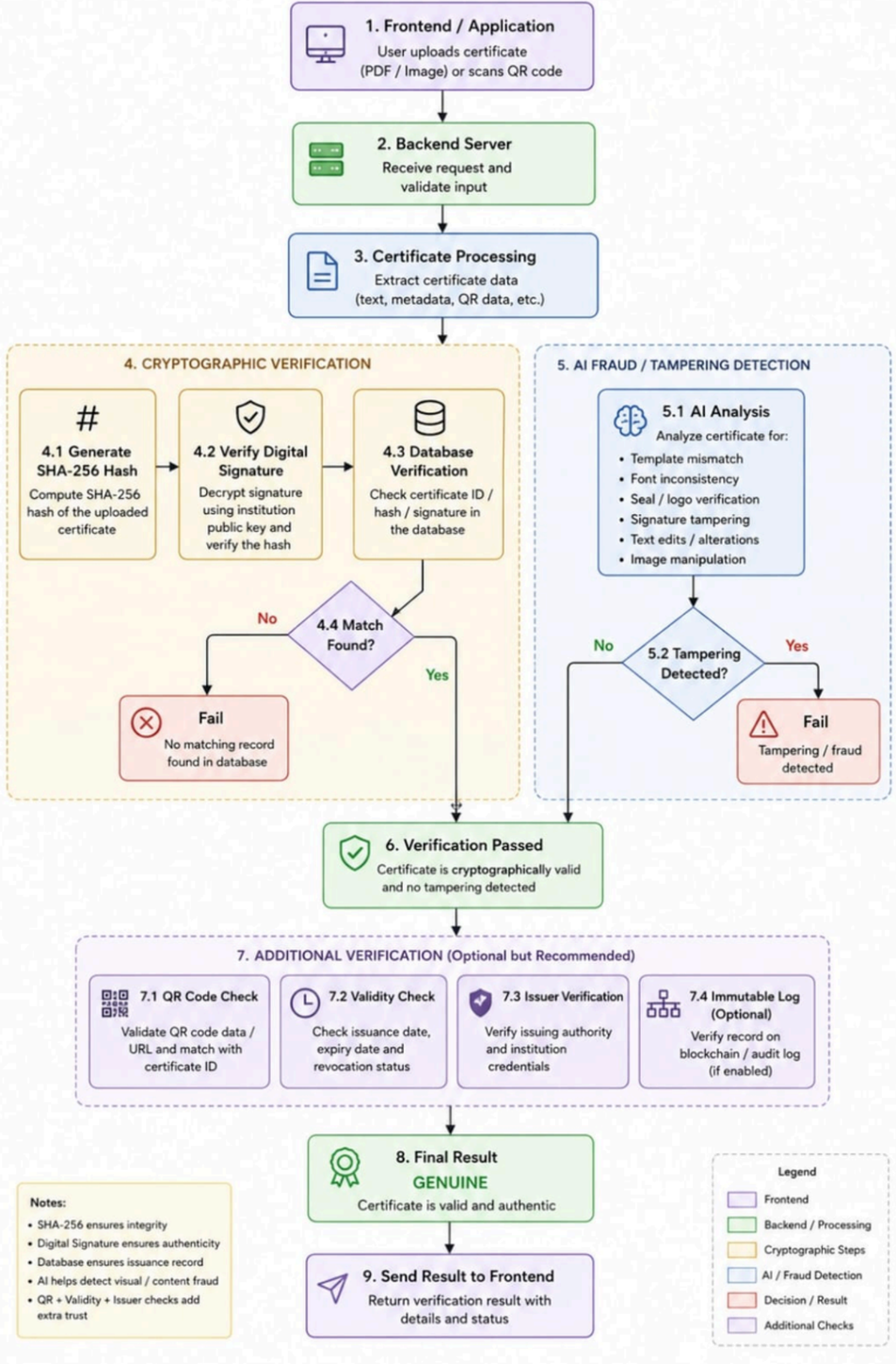
Scan QR on Certificate

↓

Instant Verdict

GENUINE | FAKE

- Real-time Python health badge
- Animated 6-check verification
- Drag-and-drop upload
- Responsive glassmorphism

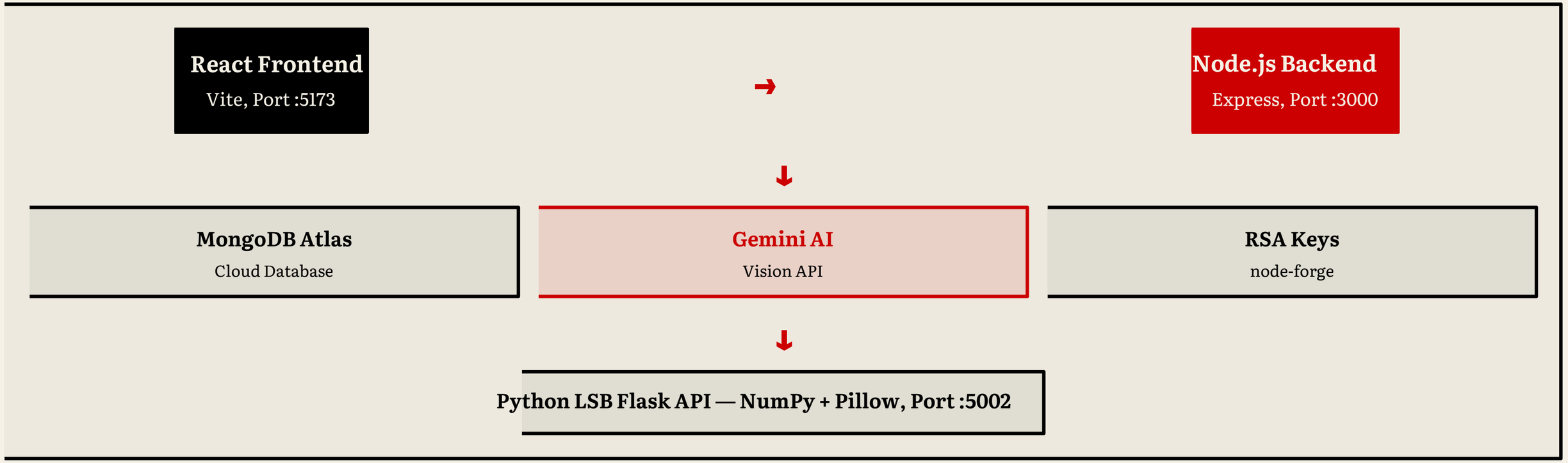

1. Frontend / Application
User uploads certificate (PDF / Image) or scans QR code
2. Backend Server
Receive request and validate input
3. Certificate Processing
Extract certificate data (text, metadata, QR data, etc.)
4. CRYPTOGRAPHIC VERIFICATION
4.1 Generate SHA-256 Hash
Compute SHA-256 hash of the uploaded certificate
4.2 Verify Digital Signature
Decrypt signature using institution public key and verify the hash
4.3 Database Verification
Check certificate ID / hash / signature in the database
4.4 Match Found?
No
Yes
Fail
No matching record found in database
5. AI FRAUD / TAMPERING DETECTION
5.1 AI Analysis
Analyze certificate for:
Template mismatch
Font inconsistency
Seal / logo verification
Signature tampering
Text edits / alterations
Image manipulation
5.2 Tampering Detected?
No
Yes
Fail
Tampering / fraud detected
6. Verification Passed
Certificate is cryptographically valid and no tampering detected
7. ADDITIONAL VERIFICATION (Optional but Recommended)
7.1 QR Code Check
Validate QR code data / URL and match with certificate ID
7.2 Validity Check
Check issuance date, expiry date and revocation status
7.3 Issuer Verification
Verify issuing authority and institution credentials
7.4 Immutable Log (Optional)
Verify record on blockchain / audit log (if enabled)
8. Final Result
GENUINE
Certificate is valid and authentic
9. Send Result to Frontend
Return verification result with details and status
Notes:
SHA-256 ensures integrity
Digital Signature ensures authenticity
Database ensures issuance record
AI helps detect visual / content fraud
QR + Validity + Issuer checks add extra trust
Legend
Frontend
Backend / Processing
Cryptographic Steps
AI / Fraud Detection
Decision / Result
Additional Checks

TECHNICAL EXCELLENCE

ARCHITECTURE & STACK

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Three-tier microservice architecture with clear separation of concerns



Steganography: Python + NumPy

Pixel array operations 10x faster than JavaScript. Custom LSB Protocol V2 with structured payload format.

Blockchain: Custom PoW (Node.js)

Auditable proof-of-work with no gas fees. Genesis block + chain validation. Immutable issuance records.

AI Analysis: Gemini 2.0 Flash

Multimodal image comparison with forensic-level pixel analysis. Returns authenticity score + difference summary.

Encryption: AES-256 + RSA-2048

Medical/legal grade standard. AES encrypts payload, RSA signs certificate. Defense-grade security.

HOW THE HASH HIDES INSIDE THE IMAGE

LSB Steganography Deep Dive — Zynex CertVerify

1. Payload Preparation

SHA-256 hash wrapped with custom Zynex markers for secure extraction.

2. Binary Conversion

Payload converted into ASCII binary bits (~1,200 bits total).

3. LSB Embedding

1 payload bit hidden inside each Red-channel pixel LSB.

4. PNG Preservation

Saved as PNG because JPEG destroys LSB data during compression.

5. Verification

Extracted hash compared against MongoDB certificate records.

- **Only Red channel is used (1 bit per pixel)**
- **Invisible change: ±1 out of 255**
- **Any image edit breaks verification instantly**

PIXEL-LEVEL HASH EMBEDDING

BEFORE EMBEDDING

Pixel[0] Red = 217 → 11011001
Pixel[1] Red = 104 → 01101000
Pixel[2] Red = 255 → 11111111

Payload Bits = 0, 0, 1

AFTER EMBEDDING

Pixel[0] Red = 216 → 11011000
Pixel[1] Red = 104 → 01101000
Pixel[2] Red = 255 → 11111111

↓ Saved as PNG (Lossless)
↓ Red-channel LSBs extracted
↓ SHA-256 hash recovered
↓ Compared with MongoDB record

Tampering any pixel causes immediate hash mismatch

INNOVATION & DIFFERENTIATION

WHAT MAKES ZYNEX UNIQUE

Feature	ZYNEX	DigiLocker	LinkedIn Certs	Manual
LSB Steganography (invisible payload)	✓	✗	✗	✗
Gemini AI pixel-level forensics	✓	✗	✗	✗
Proof-of-work blockchain ledger	✓	✗	✗	✗
6-layer simultaneous tamper detection	✓	✗	✗	✗
Employer self-service verification	✓	△	✓	✗
Works on any image (no QR dependency)	✓	✗	✗	✗

01

Invisible Payload

Image looks identical before and after steganography

02

Cross-Layer Correlation

Pixel change breaks hash AND triggers AI detection

03

Microservice Separation

Python for pixels, Node.js for business logic

04

Full Audit Trail

Every LSB operation logged with student + issuer refs

05

AI Forensic Report

Gemini returns authenticity score + difference summary

SCALABILITY & DEPLOYMENT READINESS

PRODUCTION-READY

Current infrastructure designed for horizontal scaling

Current Infrastructure Stack

MongoDB Atlas	Stateless Node.js	Python LSB Service	Vite Build
Cloud, auto-scaling	Horizontally scalable	Container-deployable	Vercel/Netlify/AWS S3

Scale Requirement	Solution
10,000 certs/day	MongoDB read replicas + Redis cache for high-throughput verification
Multi-institution	Role-based tenancy on User model with institution isolation
Government-grade	Swap to Hyperledger Fabric blockchain for enterprise consensus
Mobile app	REST APIs are JSON-native, mobile-ready with React Native
Offline verification	Export QR + LSB hash for offline cryptographic check

✅ **COMPLETED**

JWT 7-day expiry | bcrypt 12 rounds | 10MB file limits | CORS per environment | .env secrets | MongoDB TLS

⬜ **PENDING**

Rate limiting | HTTPS/SSL (reverse proxy layer — nginx/Cloudflare)

RELIABILITY & ROBUSTNESS

7 FAILURE SCENARIOS HANDLED

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Failure Point	Scenario	Our Handling
Python LSB Server	Embed operation fails	Health badge warns user; operation gracefully skipped, other layers continue
Gemini API Quota	AI analysis fails	5/6 checks still complete; AI shown as N/A with explanation
MongoDB Connection	Database unavailable	Structured JSON error response from Express middleware
Image Too Small	Embed rejected	Python returns pixel count vs. requirement with clear error message
JWT Token Expired	Authentication fails	Auto-redirect to login page; token cleared from storage
Malformed Upload	Bad file format	Multer rejects with clear error message; no server crash
No LSB Payload	Extract fails	422 response — not a server crash; indicates non-Zynex certificate

✅ TESTED SCENARIOS

- ✓ **Full lifecycle: Issue → LSB → Blockchain → Verify → AI**
- ✓ **Tampered certificate → FAKE verdict confirmed**
- ✓ **Revoked certificate → blocked at verification**
- ✓ **Python restart → Node.js bridge recovers gracefully**

THANK YOU

ZYNEX CertVerify

"The last line of defence against certificate fraud"

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Every certificate.
Cryptographically sealed.

Every verification.
AI-powered.

Every fraud attempt.
Detected.